Experiment 2: The Superposition Theorem

OBJECTIVE:

1. To construct a circuit with two independent voltage sources.
2. To apply and verify the superposition theorem to linear circuit.
3. To construct a superposition theorem circuit with independent and dependent sources.

INTRODUCTION:

Superposition Theorem

Superposition is a method to determine the voltages and currents in a circuit that has many sources. It allows the calculation of the combined effect of many sources by looking at the individual effects of each source acting alone. Each source is treated one at a time, as if it were the only ACTIVE source in the circuit. All the other sources are considered NOT ACTIVE; whereby voltage, sources are replaced by short circuits, and current sources are replaced by open circuits.

The voltage or current can be calculated by taking the components due to the first source acting alone. The calculations are repeated for each source in the circuit. When all the sources have been considered, the overall voltage or current is the algebraic sum of the individual voltage or current. The superposition theorem will work for any number of sources as long as you are consistent in accounting for the direction of currents and the polarity of voltages.

Steps to apply superposition principle

1. Turn off all independent sources except one source. Find the output (voltage or current) due to that active source using nodal or mesh analysis.
2. Repeat step 1 for each of the other independent sources.
3. Find the total contribution by adding algebraically all the contributions due to the independent sources.

Two things have to be kept in mind:

1. When we say turn off all other independent sources:
   - Independent voltage sources are replaced by

   0 V (short circuit) and
   - Independent current sources are replaced by

   0 A (open circuit).
2. Dependent sources are left intact because they are controlled by circuit variables.

EQUIPMENT AND COMPONENT

1. Independent source, DC Power supplies, 10V, 15V, 30V
2. Independent source, DC Current, 6A
3. Dependent source, Current controlled current source (CCCS), gain = 4 A/A
4. Multimeter (Voltmeter, Ammeter)
5. Resistor, 10Ω, 20Ω, 40Ω, 1kΩ, 2.2 kΩ, 3.9 kΩ.

PROCEDURE

Part A: Measurement of currents and voltage in linear circuit with independent sources



Figure 1: Multisim 14.0 software icon

1. Run Multisim 14.0 Software.
2. Click File > Save As > COM106_Experiment 2_YOUR ID_YOURNAME > Save.
3. Follow Step 4 until Step 18 to construct your circuit as shown in figure 2.

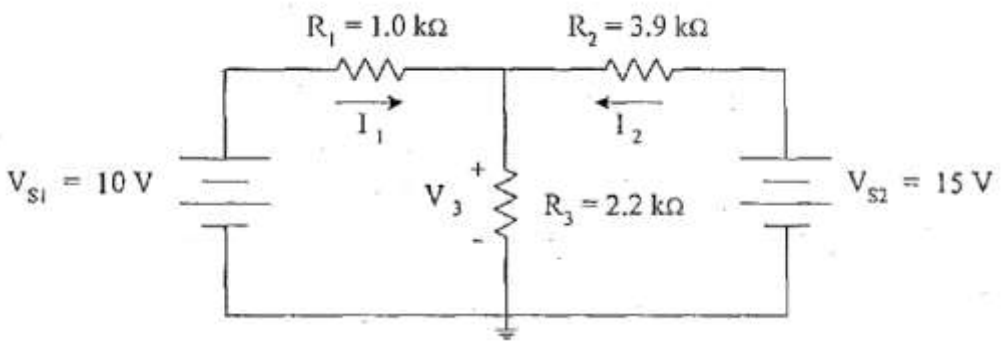


Figure 2: Circuit with independent sources

4. Click Place > Component > Group > Sources > POWER_SOURCES > DC_POWER > OK. Place it on the screen.
5. Set your DC_POWER to 10 V for Vs1. Double click on DC_POWER icon > Value > Voltage > 10V > OK.
6. Set your DC_POWER to 15 V for Vs2. Double click on DC_POWER icon > Value > Voltage > 15V > OK.
7. Click Place > Component > Group > Basic > RESISTOR > OK. Place it on the screen.
8. Set your RESISTOR value to 1kΩ for R_1. Double click on RESISTOR > Value > Resistance > 1kΩ > OK.
9. To rotate the RESISTOR > Click RESISTOR > Ctrl + R **or** Right click > Rotate 90^0 Clockwise/Counterclockwise.
10. Repeat step 7 until step 9 to set complete RESISTOR in the circuit.
11. Click Place > Component > Group > Sources > POWER_SOURCES > GROUND > OK. Place it on the screen.
12. Connect all component as shown in Figure 2.
13. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in series connection with the circuit for current measurement as shown in figure 3 below.
14. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in parallel connection with the circuit for voltage measurement across R_3 as shown in figure 3 below.

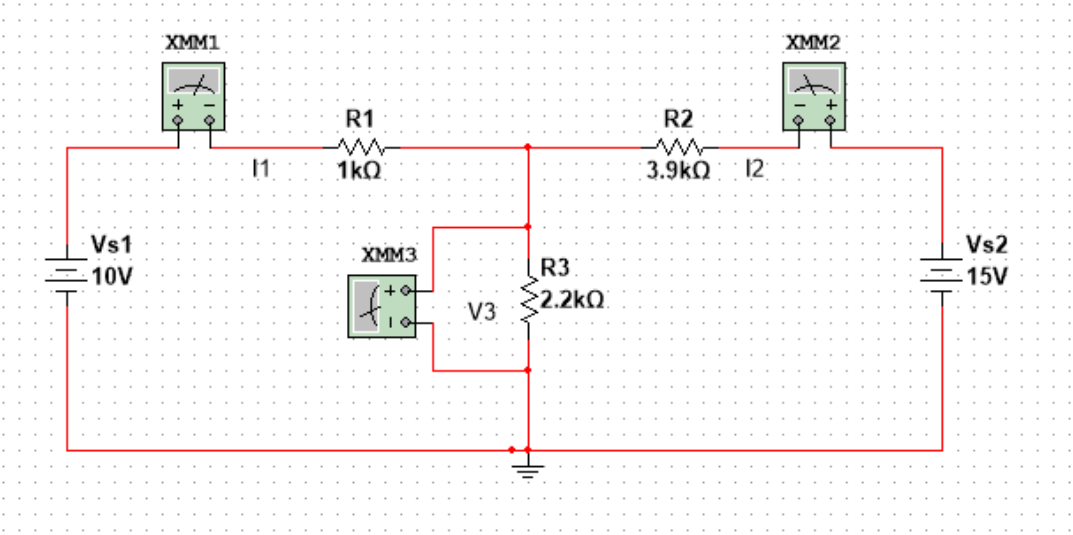


Figure 3: Circuit with Ammeter and Voltmeter

15. Double click Multimeter > Choose A for current measurement > Straight line for DC current.



Figure 4: DC Ammeter setting

16. Double click Multimeter > Choose V for voltage measurement > Click straight line for Dc voltage measurement.



Figure 5: DC Voltmeter setting

17. Run the simulation

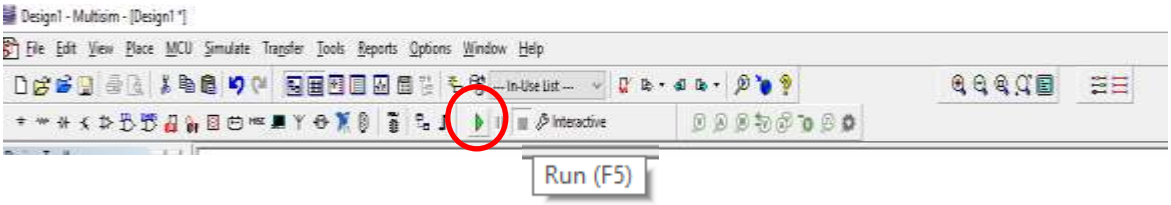


Figure 6: Run button/Icon

18. Measured and record the Ammeter and Voltmeter reading in Table 1.

Part B: Applying Superposition Theorem with independent sources

1. By using the same circuit in Part A, switch off the power supplies and remove one independent 15V voltage source by make it as short circuit.
2. The 10V, voltage source is made ACTIVE by replacing the 15V voltage source with a jumper (short circuit) between the point labeled as C and D, as shown in figure 7.

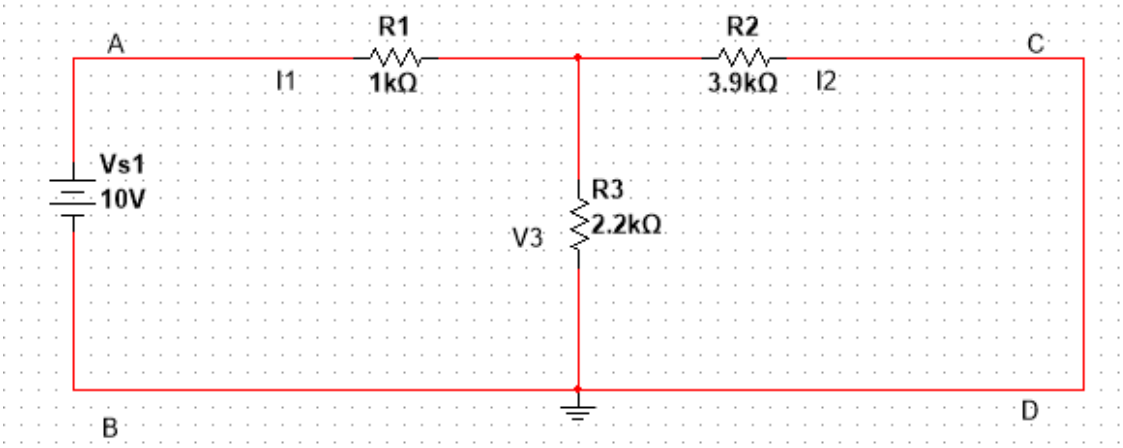


Figure 7: Turn off voltage source, 15V by short circuit.

3. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in series connection with the circuit for current measurement as shown in figure 8 below.
4. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in parallel connection with the circuit for voltage measurement across R_3 as shown in figure 8 below.

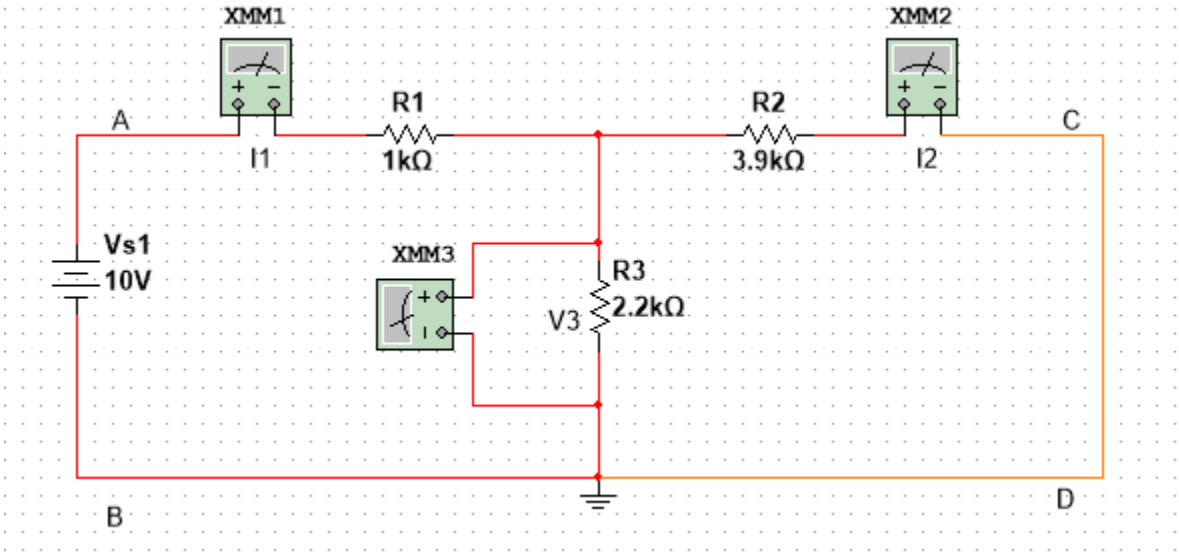


Figure 8: Multimeter connection with Turn off voltage source, 15V by short circuit

5. Run the simulation. Measured and record the Ammeter and Voltmeter reading in Table 2.1.

6. Connect circuit as figure 9 below. The 15 V voltage source is made ACTIVE by replacing the 10 V voltage source with a jumper between the points labeled A and B.

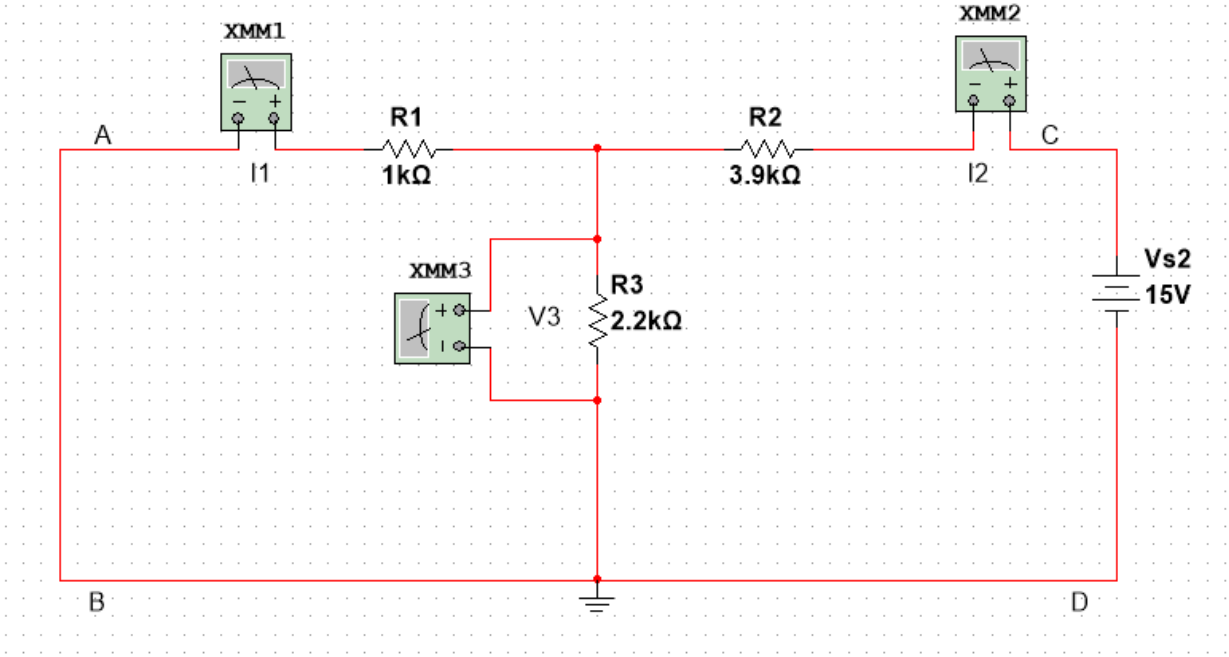


Figure 9: Multimeter connection with Turn off voltage source, 10V by short circuit

7. Run the simulation. Measured and record the Ammeter and Voltmeter reading in Table 2.2.
8. Calculate the algebraic sum of currents I_1, I_2 and voltage V_3 in Table 2.3. Pay attention the direction of the currents. Current in the same directions are added and currents in the opposite directions arc subtracted.

Part C: Superposition Theorem with independent source and dependent source

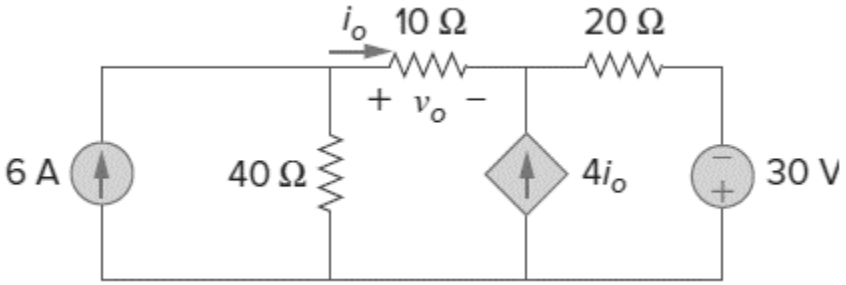


Figure 10: Original circuit with independent and dependent source

1. Construct below circuit by Turn off independent voltage source, 30V by make it as short circuit as in figure 11. Do not disturb the dependent sources because they are controlled by circuit variables.

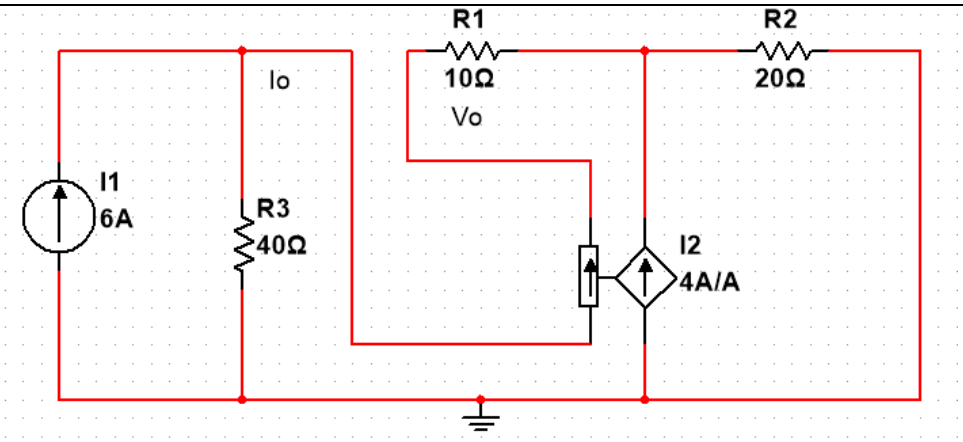


Figure 11: Turn off independent voltage source, 30V by short circuit

2. Click Place > Component > Group > Sources >SIGNAL_CURRENT_SOURCES>DC_CURRENT > OK. Place it on the screen
3. Set your DC_Current to 6A. Double click on DC_Current icon > Value > Current > 6A > OK.
4. Click Place > Component > Group > Basic > RESISTOR > OK. Place it on the screen.
5. Set your RESISTOR value to 10Ω. Double click on RESISTOR > Value > Resistance > 10Ω > OK.
6. Repeat step 4 until step 5 to set complete RESISTOR in the circuit.
7. Set your dependent source gain to 4 A/A. Click Place > Component > Group > Sources > CONTROLLED_CURRENT_SOURCES>CURRENT_CONTROLLED_CURRENT_SOURCE> OK. Place it on the screen.
8. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in parallel connection to measure voltage,Vo across 10Ω as in figure 12.
9. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in series connection to measure current flow through 10Ω as in figure 12.

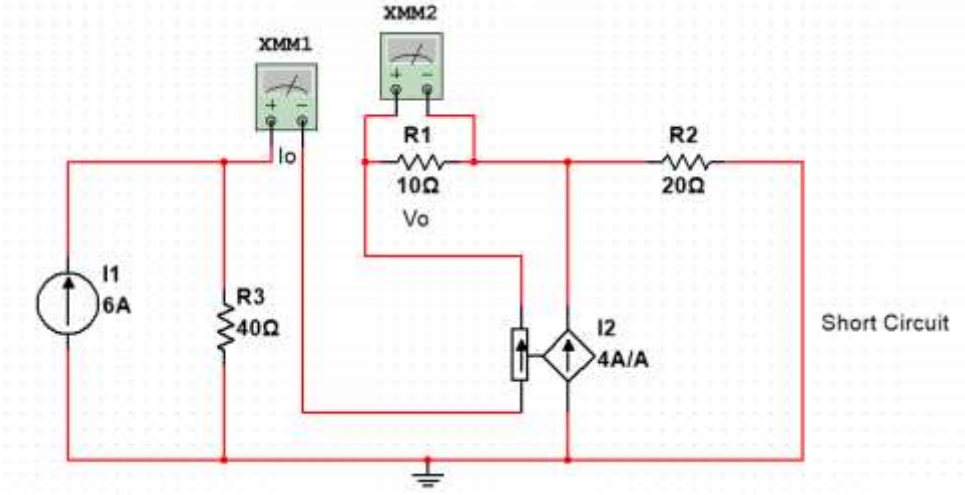


Figure 12: Turn Off independent voltage source, 30V by short circuit

10. Connect all component as shown in Figure 12.
11. Run the simulation. Measured and record Vo and Io, in Table 3.1.
12. Turn off independent current source, 6A by open circuit. Make independent voltage source, 30V as ACTIVE mode as shown in figure 13.

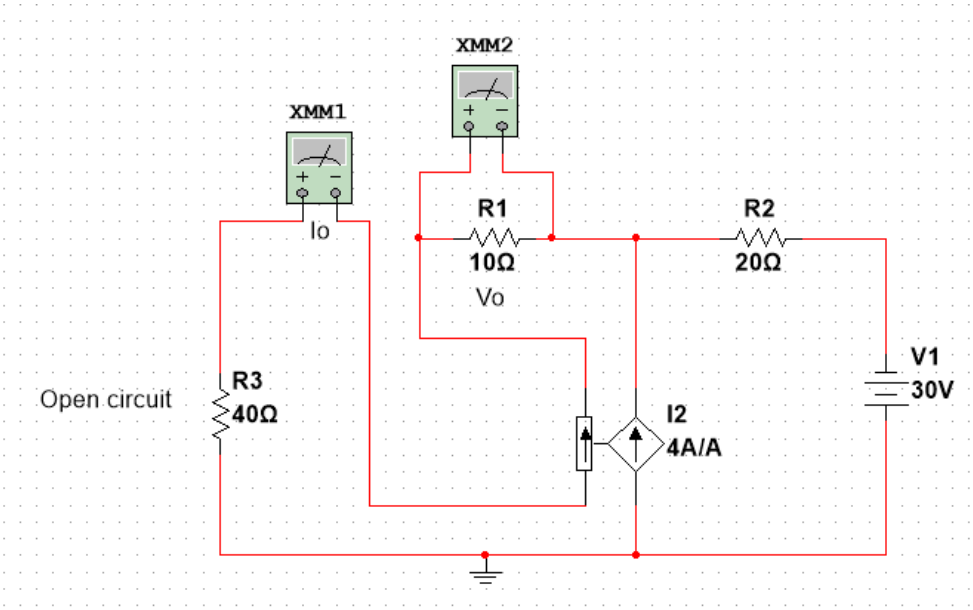


Figure 13: Turn Off independent current source, 6A by open circuit

13. Connect all component as shown in Figure 13.
14. Run the simulation. Measured and record Vo and Io, in Table 3.2.

9. Calculate the algebraic sum of currents I_o, and Voltage, V_o in Table 3.3. Pay attention the direction of the currents. Current in the same directions are added and currents in the opposite directions arc subtracted.

RESULT PART A

Part A: Measurement of currents and voltage in circuit with independent sources

Table 1: Currents and voltage measurement of circuit with independent sources

Parameter	Measured value	Calculated value
I_1 (mA)		
I_2 (mA)		
V_3 (V)		

RESULT PART B

Part B: Applying Superposition Theorem with independent sources

Table 2.1: Superposition Theorem measurement of circuit with independent sources by turn off voltage source, 15V

Parameter	Measured value	Calculated value
$I_{1\ (15V\ Off)}$ (mA)		
$I_{2\ (15V\ Off)}$ (mA)		
$V_{3\ (15V\ Off)}$ (V)		

Table 2.2: Superposition Theorem measurement of circuit with independent sources by turn off voltage source, 10V

Parameter	Measured value	Calculated value
$I_{1\ (10V\ Off)}$ (mA)		
$I_{2\ (10V\ Off)}$ (mA)		
$V_{3\ (10V\ Off)}$ (V)		

Table 2.3: Superposition Theorem measurement of circuit with independent sources

Parameter	Measured value	Calculated value
$I_{1\ (total)}$ (mA)		
$I_{2\ (total)}$ (mA)		
$V_{3\ (total)}$ (V)		

RESULT PART C

Part C: Superposition Theorem circuit with independent and dependent source

Table 3.1: Superposition Theorem with independent and dependent source by turn off voltage source, 30V

Parameter	Measured value	Calculated value
$V_{o\ (30V\ off)}$ (V)		
$I_{o\ (30V\ off)}$ (A)		

Table 3.2: Superposition Theorem with independent and dependent source by turn off current source, 6A

Parameter	Measured value	Calculated value
$V_{o\ (6A\ off)}$ (V)		
$I_{o\ (6A\ off)}$ (A)		

Table 3.3: Superposition Theorem with independent and dependent source

Parameter	Measured value	Calculated value
$V_{o\ (total)}$ (V)		
$I_{o\ (total)}$ (A)		

DISCUSSION

The following points should be discuss in your report.

1. Analyze your result and verify the Superposition theorem.
2. Discuss briefly the importance of Superposition Theorem in circuit analysis.
3. Show your working calculation for Table 1, 2.1, 2.2, 2.3, 3.1, 3.2 and 3.3.